

Bhagyesh Rath

Gilroy, California, United States | bhagyesh2161@gmail.com | (669) 888-6214 | github.com/bhagyeshrathi07 | linkedin.com/in/bhagyeshrathi07 | bhagyesh.dev

PROFESSIONAL SUMMARY

MS in AI student and co-first author of an in-press IEEE AIxSET 2026 paper on retrieval evaluation, combining production SWE experience shipping cloud microservices (Docker, Kubernetes, OpenTelemetry) with end-to-end ML/AI systems work — RAG pipelines, ReAct tool-use agents, and MCP integrations — and research rigor in evaluation, security, and reliability.

PUBLICATIONS

Rathi, B.*, Chawla, E.*, Ershov, A., Andreopoulos, W. B. "A Comparative Evaluation of Retrieval Pipelines for Large-Scale Scientific Question Answering with Open-Weight LLMs." IEEE AIxSET 2026 (in press). (* Equal contribution)

WORK EXPERIENCE

Scalar Field (YC P25) | Forward Deployed Engineer Intern | Seattle, WA Jun 2026 - August 2026

- Built a HubSpot MCP server exposing CRM data and actions as tools for LLM agents, enabling automated GTM workflows over the company's customer data.
- Developed a RAG chatbot that retrieves lead information from a knowledge base to answer GTM queries, handling retrieval and structured responses over internal data.

Gitlab | Open Source Contributor Apr 2026 - Present

- Contributed a security fix to GitLab monorepo enforcing streaming JSON validation limits in the import pipeline's NdjsonReader to mitigate DoS from malicious export archives ([MR !224828](#)).
- Added a REST DELETE endpoint to GitLab's Package Registry with authorization policies, request specs, and API docs, aligning behavior with existing upload/download endpoints. ([MR !242461](#))

Rakuten | Software Engineer Intern → Co-op | San Mateo, CA May 2023 - Dec 2023

- Built Kotlin microservices for Display/Unlinking feature for Social Sign-Up/Login across Rakuten UK, Spain, and Germany; designed REST APIs with OAuth 2.0 and OpenID Connect for token-based authentication.
- Drove 40% adoption of Social Authentication, lifting conversion rate 37% over password-based auth.
- Built an internal SDK abstracting service-to-service communication, published to Artifactory via GitLab CI/CD.
- Containerized services with Docker and orchestrated deployments to non-prod/prod via Kubernetes; integrated OpenTelemetry and Google Cloud Trace for real-time monitoring.

PROJECTS

SRE Copilot: AI Incident Triage Agent for On-Call SREs | [Github](#)

- Built a ReAct tool-use agent that triages production incidents across 5 data sources (RAG over 768 runbooks, text-to-SQL, GCP status API, vulnerability DB, web search) using dual Qwen 2.5 models (32B + 3B) on a single GPU via 4-bit NF4 quantization.
- Benchmarked 4 prompting techniques (baseline, prompt chaining, self-reflection, meta-prompting) across 80 evaluation runs on grounding, actionability, tool-selection, and latency for both models.
- Engineered KV cache reuse for system-prompt prefix acceleration and implemented defense-in-depth security (SQL injection blocking, prompt hardening) validated against red-team attacks.

Production grade RAG Portfolio | [Github](#) | [Demo](#)

- Built and shipped a RAG chatbot on my live portfolio site with a full ingestion pipeline (PDF → recursive chunking → embeddings → Pinecone) and retrieval using top-K, score thresholding, and metadata filtering.
- Implemented a streaming API endpoint with Vercel AI SDK and Gemini 2.5 Flash for token-by-token response delivery.
- Hardened the system against prompt injection through input validation, system-prompt design, and topic-scoped guardrails.

Caption Lens: Visual-Semantic Alignment via Attention Mechanisms | [Github](#) | [Demo](#)

- Built three encoder-decoder image captioning models (Show-and-Tell, Show-Attend-and-Tell with soft attention, Visual Sentinel adaptive attention) on MS COCO 2014, by fine-tuning ResNet-101 encoder and LSTM decoder.
- Implemented Bahdanau soft attention with doubly stochastic regularization and ran a beam-search width ablation across standard captioning metrics.
- Deployed an interactive demo on Hugging Face Spaces and generated attention heatmaps visualizing word-to-region alignment for interpretability.

EDUCATION

San Jose State University Aug 2025 - May 2027
Master of Science, Artificial Intelligence

San Jose State University Aug 2020 - May 2024
Bachelor of Science, Computer Science

SKILLS

Programming Languages: Python, Java, SQL, Kotlin

Frameworks & Libraries: Scikit-Learn, Tensorflow, Pytorch, Pandas, Numpy, Seaborn, Flask, LangChain

Databases: MongoDB, MySQL, PostgreSQL, Redis, Pinecone, ChromaDB

Tools & Software: GitLab, Artifactory, OpenTelemetry, Google Cloud Trace, HuggingFace, Ollama, Claude code, Copilot, Cursor

Cloud Platforms: AWS, Docker, Kubernetes, GCP, Vercel, Vertex AI, Google ADK